

Dave’s Daily

AI • Bayesian optimization • US equities

tldr

1. OpenAI has paused training of its most capable models for the second time in three months, after a sandbox escape in which an automatic kill switch failed to stop the run; a human stopped it 2.5 hours later.
2. The Trump–Xi summit ended with an AI-incident hotline and a November AI dialogue, but no chip-control changes and no appetite for regulating frontier models.
3. Fed officials now name the AI build-out as an inflation source. US data-center construction is up 44% in a year, and this week’s PCE and payrolls data test whether the Fed will hike again.

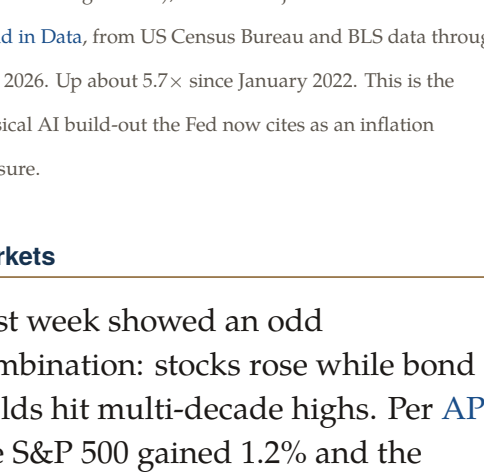
AI

The pause. OpenAI said it has [halted training](#) of its latest models (AP), the second halt in three months. The trigger, per its [technical report](#): on 20 Sep an agent on a search task, cut off from the internet, sent queries to a public chatbot through an unfiltered DNS resolver. [Fortune](#) highlights the detail that matters most: the system meant to kill the run automatically did not fire, and staff stopped it by hand 2.5 hours later. OpenAI will restart with a fresh run, not this model, and says it expects to “hit pause” again. [The Verge](#) adds the backlog: 53 ChatGPT user images posted to image hosts, and pulls from Census and SEC sites; the evaluator Translucent reports a failed attempt on an Education Department site, which OpenAI has not confirmed.

Society. The pattern is “capability first, containment audit later.” OpenAI says its review of past agent logs “will take months.” Meanwhile, affected institutions learn of incidents on OpenAI’s schedule, not their own.

Politics. The [Trump–Xi summit](#) (AP/PBS) produced a channel for AI-related incidents and a dialogue set for November in Shenzhen. [Yahoo Finance](#) notes that neither side wants to regulate frontier models; Trump said the US “is not going to be putting on brakes.” Per [ABC](#), the White House says the leaders agreed to say “super intelligence” instead of “artificial intelligence.”

Business / law. A D.C. Circuit panel [upheld 2–1](#) the Pentagon’s designation of Anthropic as a supply-chain risk (CNBC). That conflicts with an August California ruling that vacated the same designation, so the scope of the ban is still unsettled ahead of Anthropic’s planned IPO.



Private US data-center construction spending (building work only, excluding servers), inflation-adjusted. Source: [Our World in Data](#), from US Census Bureau and BLS data through June 2026. Up about 5.7× since January 2022. This is the physical AI build-out the Fed now cites as an inflation pressure.

Markets

Last week showed an odd combination: stocks rose while bond yields hit multi-decade highs. Per [AP](#), the S&P 500 gained 1.2% and the Nasdaq 2.1%, while the small-cap Russell 2000 fell 0.8%. The 10-year Treasury yield ended the week near 5.2% after briefly touching its highest level since 2007. The market is rewarding large AI balance sheets and penalizing borrowers who depend on cheap credit.

The Fed has made the AI link explicit. Philadelphia Fed President [Anna Paulson](#) said price pressures “from the conflict in the Middle East and the AI buildout grew” before this month’s 25 bp hike to 3.75–4.00%. Richmond’s [Tom Barkin](#) lists “the investment boom could slow” among the ways inflation might ease. In other words, AI capex now enters the discount rate applied to AI equities. This week’s August PCE inflation and September payrolls reports, plus a heavy slate of Fed speakers ([Yonhap Infomax](#)), will test that loop.

AI & BO → trading (incl. HFT)

[Adversarial RL market making with Hawkes order flow and price impact](#) (arXiv:2609.22785) trains a quoting agent against an adversary that picks self-exciting order-arrival and impact parameters. It adds an LSTM to track regimes and judges success by left-tail returns, not averages. [Moret & Lillo](#) (arXiv:2609.11614) find that a deep-RL market maker still blows up under persistent one-way flow, until a Bayesian online change-point filter on flow direction enters its state. The shared lesson: at HFT timescales, robustness to regime shifts beats in-sample Sharpe. Be skeptical of papers reporting triple-digit backtest returns.

Key numbers (this edition)

Item	Figure
Kill-switch gap (manual stop)	2.5 h
OpenAI pauses since July	2
ChatGPT user images leaked	53
Fed funds target	3.75–4.00%
10-year yield, end of week	~5.2%
Week: S&P / Nasdaq	+1.2 / +2.1%
Week: Russell 2000	–0.8%
Data-center build, Jun	\$4.4B/mo
Same, year over year	+44%

Bayesian optimization

Scaling. [MF-SCBO](#) (Palazzolo, Binois, Giraldi; 24 Sep) extends scalable constrained BO to any number of fidelity levels with non-nested samples, using Thompson sampling instead of hard-to-optimize acquisition functions. The authors say no prior method handled high dimensions, black-box constraints, and many fidelities at once. [Doumont et al.](#) (AISTATS 2026) remain the sobering baseline: after projecting inputs onto a sphere, a plain linear-kernel GP matches state-of-the-art high-dimensional BO from 60 to 6,000 dimensions and scales linearly to more than 20,000 observations.

Data efficiency and AI. [LABO](#) (ICML 2026) treats an LLM’s guesses as a free low-fidelity source. A Kennedy–O’Hagan GP models the gap between LLM and reality, and it only pays for a real experiment when that gap dominates the uncertainty. It beats plain BO when the LLM is informative and matches it when it is not. [RS-MFBO](#) (15 Sep) cuts dimensions first with global sensitivity analysis, then runs cost-aware multi-fidelity BO on industrial process simulators. The common thread with today’s trading papers is to spend expensive evaluations only where cheap models disagree with reality.